

Open multi-source inputs

Flood labels
(DEP stormwater, 311, USGS Ida HWM)

Static predictors
(terrain, flow-accumulation proxy, land cover,
buildings, hydrologic proximity)

Rainfall condition r
(constant synthetic; not radar)

FEMA Sandy negative control
(never a training label)

H3 assembly (R9)

Join layers to H3 cells

Provenance tags
(assembly · feature · label · rainfall)

Learning & blocked evaluation

Gradient-boosting classifier
+ continuous-risk regressor

H3-block GroupKFold spatial CV
(R7 parent blocks)

Logistic, ponding & constant-class baselines

Diagnostics & outputs

$PFI_h(c,r)$

Scale-loss Jaccard ladder
(R10 $\rightarrow$ R9 / R8)

Adaptive refinement
(PFI_h -guided $\rightarrow$ R11)

Sandy coastal-overlap diagnostic